

Apply estimation, rounding and bounds

Use approximation and limits of accuracy to judge answers sensibly.

Step 48

★ Today I am learning to apply estimation, rounding and bounds.

1. Estimation

Estimate each answer by rounding to 1 significant figure.

- | | | |
|------------------------------|----------------------------------|-----------------------------------|
| a) $298 + 412 \approx$ _____ | e) $397 \times 21 \approx$ _____ | i) $3.7 + 4.2 \approx$ _____ |
| b) $523 - 198 \approx$ _____ | f) $608 \times 72 \approx$ _____ | j) $9.8 - 2.6 \approx$ _____ |
| c) $689 + 315 \approx$ _____ | g) $4,812 \div 19 \approx$ _____ | k) $6.3 \times 3.1 \approx$ _____ |
| d) $721 - 299 \approx$ _____ | h) $7,351 \div 28 \approx$ _____ | l) $14.9 \div 4.2 \approx$ _____ |

Remember:

- To estimate, round each number to 1 significant figure (or as directed).
- Then calculate using the rounded numbers.
- The answer will be an approximation.

2. Rounding

Round each number to the given place.

- | | | |
|---------------------------------------|--|---|
| a) 4,286 to the nearest 10 = _____ | f) 62,847 to the nearest 1,000 = _____ | k) 0.0956 to 2 decimal places = _____ |
| b) 4,286 to the nearest 100 = _____ | g) 3.764 to the nearest whole number = _____ | l) 12.678 to 1 decimal place = _____ |
| c) 4,286 to the nearest 1,000 = _____ | h) 3.764 to 1 decimal place = _____ | m) 12.678 to 2 decimal places = _____ |
| d) 62,847 to the nearest 10 = _____ | i) 3.764 to 2 decimal places = _____ | n) 12.678 to the nearest whole number = _____ |
| e) 62,847 to the nearest 100 = _____ | j) 0.0956 to 1 decimal place = _____ | o) 987.65 to the nearest 10 = _____ |

Remember:

- Look at the digit to the right of the place you are rounding to.
- 5 or more → round up
- Less than 5 → round down

3. Bounds (Whole Numbers)

Write the lower and upper bounds for each number, rounded to the given place.

- | | |
|---|---|
| a) 237 to the nearest 10
Lower bound: _____ Upper bound: _____ | d) 6,542 to the nearest 100
Lower bound: _____ Upper bound: _____ |
| b) 237 to the nearest 100
Lower bound: _____ Upper bound: _____ | e) 12,300 to the nearest 1,000
Lower bound: _____ Upper bound: _____ |
| c) 6,542 to the nearest 10
Lower bound: _____ Upper bound: _____ | f) 91,999 to the nearest 1,000
Lower bound: _____ Upper bound: _____ |

Remember:

- The lower bound is the smallest possible number that rounds up to the given value.
- The upper bound is the largest possible number that rounds down to the given value.
- For whole numbers, the lower bound is included and the upper bound is not included.

4. Bounds (Decimal Numbers)

Write the lower and upper bounds for each number, rounded to the given place.

- | | |
|---|---|
| a) 3.6 to the nearest whole number
Lower bound: _____ Upper bound: _____ | d) 3.68 to 2 decimal places
Lower bound: _____ Upper bound: _____ |
| b) 3.6 to 1 decimal place
Lower bound: _____ Upper bound: _____ | e) 12.34 to 1 decimal place
Lower bound: _____ Upper bound: _____ |
| c) 3.68 to 1 decimal place
Lower bound: _____ Upper bound: _____ | f) 0.075 to 2 decimal places
Lower bound: _____ Upper bound: _____ |

Remember:

- For decimals, the lower bound is included and the upper bound is not included.
- Think about the value of the next digit.
- If rounding to 1 decimal place, use steps of 0.1.
- If rounding to 2 decimal places, use steps of 0.01.

5. Use Estimation, Rounding and Bounds

Solve the problems.

- | | |
|--|---|
| a) A shop sells a box of 48 biscuits for \$2.97. About how much would 7 boxes cost? Estimate: \$ _____ | d) A tank contains 263.4 litres of water. What are the lower and upper bounds to 1 decimal place? Lower: _____ Upper: _____ |
| b) A runner completed a race in 56.78 minutes. What is the time rounded to the nearest minute? _____ | e) The length of a table is 1.852 metres. Rounded to 2 decimal places, what is the length? _____ |
| c) A book is 198 pages long. What are the lower and upper bounds when rounded to the nearest 10 pages? Lower: _____ Upper: _____ | f) Estimate: $498 \div 19$
Estimated answer: _____ |

Tips for Success:

- ✓ Estimate first to check your answer is reasonable.
- ✓ Round carefully and write the place value.
- ✓ Use bounds to show the range of possible values.
- ✓ Always include units in word problems.

6. Problem Solving

Answer the questions.

- | | | |
|--|--|--|
| a) A family drives 248 km to a holiday destination. The sign on the road says another 136 km to go. About how many kilometres is that in total? Estimate: _____ km | c) A packet of pasta weighs 495 g. What are the lower and upper bounds when rounded to the nearest 10 g? Lower: _____ g Upper: _____ g | e) A shop sells a water bottle for \$1.49. About how much would 6 bottles cost? Estimate: \$ _____ |
| b) A recipe needs 0.375 kg of sugar. What is the amount rounded to 2 decimal places? _____ kg | d) A piece of rope is 23.456 metres long. What are the lower and upper bounds to 2 decimal places? Lower: _____ m Upper: _____ m | f) The population of a town is 8,752 people. What are the lower and upper bounds when rounded to the nearest 100 people? Lower: _____ people Upper: _____ people |

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more help.



This was hard.

Parent Observation (optional)

Please tick the boxes that apply.

- | | |
|---|---|
| ☐ Understands estimation | ☐ Applies to word problems |
| ☐ Rounds accurately | ☐ Works neatly and carefully |
| ☐ Finds lower and upper bounds | ☐ Needs more practice |

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____